1.1 Project Proposal

- **Project Name:** EduHub – Enterprise-Grade AI-Enhanced Online Learning Platform
- **Domain:** Educational Technology (EdTech) & E-Learning Solutions
- **Executive Summary:** EduHub is an enterprise-scale online learning management platform engineered to deliver end-to-end course creation, structured e-learning, automated assessments, and verified credentialing. Built on ASP.NET Core MVC, Entity Framework Core, and modern enterprise design patterns, EduHub bridges the gap between commercial systems (like Udemy and Coursera) and institutional needs by combining multi-tier architecture, cloud-native media streaming via Azure Blob Storage, resilient payments via Stripe, AI-assisted learning enhancements, and cryptographic certificate verification.
- **Core Objectives:**
    - Engineer a loosely coupled, highly maintainable solution utilizing an N-Tier Architectural Pattern and Domain-Driven Design (DDD) principles.
    - Guarantee transactional consistency across financial transactions and enrollment states using the Unit of Work and Repository design patterns.
    - Implement dynamic pricing, voucher strategies, and progress-driven event notifications using Strategy and Observer patterns.
    - Provide scalable media ingestion, video processing, and secure tokenized delivery through Microsoft Azure Blob Storage.
    - Incorporate intelligent AI-driven learning tools (automated lecture summarization, quiz generation, and semantic search) alongside automated PDF certificate compilation.
- **Project Scope:**
    - **In-Scope:** Multi-role identity management (Student, Instructor, Admin, System Auditor), course builder with multi-tier hierarchy (Sections, Lessons, Resources), cloud media pipelines with signed streaming URLs, multi-currency shopping cart and Stripe payment integration, interactive quiz engine with randomized question pools, automated dynamic PDF generation with public cryptographic validation, analytics dashboards with real-time telemetry, and integrated AI pedagogical assistance.

- **Out-of-Scope (Phase 1):** Full real-time video conferencing/live webinars (WebRTC/P2P live calls), reserved for secondary roadmap deployment.

1.2 Project Plan & Milestones (Structured 4-Week Schedule)

Sprint / Phase	Focus & Milestones	Key Engineering Deliverables
Week 1	**Architecture Foundation, Security & Domain Modeling**	- Multi-project solution setup following N-Tier guidelines (EduHub.Web, EduHub.Services, EduHub.Data, EduHub.Core) - Entity Framework Core Code-First configuration with Fluent API mapping - Generic and specific Repository patterns combined with Unit of Work abstraction - Identity system setup (ASP.NET Identity) featuring RBAC, claims, and JWT/Cookie auth - Catalog browsing with multi-facet filtering, full-text search, and pagination
Week 2	**Instructor Workflow, Media Ingestion & Course Builder**	- Comprehensive Instructor Management Studio and drag-and-drop course builder

Sprint / Phase	Focus & Milestones	Key Engineering Deliverables
		- Azure Blob Storage integration for secure media upload and tokenized SAS URL delivery - Video resource management, document attachment uploads, and duration calculations - Pricing strategy engine, voucher configurations, and draft/approval lifecycle states
Week 3	**Student Learning Portal, Assessment Engine & Checkout**	- Distraction-free custom video player (Video.js) with real-time timestamped note-taking - Automated lesson progress tracking and course completion percentage engine - Assessment engine supporting randomized quiz pools, passing thresholds, and retake policies

Sprint / Phase	Focus & Milestones	Key Engineering Deliverables
		- Stripe checkout integration with coupon validation using the Strategy Pattern - Interactive Q&A forums per lecture and verified student review system
Week 4	**Certifications, Analytics, AI Capabilities & Cloud Deployment**	- Automated PDF certificate compilation via iTextSharp with unique QR codes & verification hashes - Administrative analytics suite with Chart.js showing platform revenue, engagement, and drop-off rates - AI-assisted features (lecture transcript summaries and auto-generated review questions) - Automated unit/integration test suites and CI/CD pipelines via GitHub Actions

Sprint / Phase	Focus & Milestones	Key Engineering Deliverables
		- Live production deployment to Azure App Service and Azure SQL Database

1.3 Task Assignment & Team Responsibilities

Technical Role	Core Responsibilities
Software Architect & Backend Lead	Defines the solution structure, designs domain entities, enforces SOLID principles, implements design patterns (Repository, Unit of Work, Strategy, Observer), and maintains overall architectural integrity.
Cloud & Database Engineer	Manages database normalization, indexing, and EF Core migrations; configures Microsoft Azure resources (Azure SQL, Blob Storage, CDN, Key Vault); establishes automated CI/CD deployment pipelines.
Frontend & UI/UX Specialist	Builds modern, accessible, and mobile-first interfaces using Bootstrap 5, custom CSS3, and JavaScript/jQuery; integrates Video.js media players, interactive drag-and-drop elements, and Chart.js dashboards.
Security & Integration Engineer	Enforces ASP.NET Identity authentication, anti-forgery guards, and role-based authorization; coordinates Stripe payment gateway workflows; oversees secure cryptographic certificate generation and third-party API hooks.

1.4 Risk Management & Mitigation Matrix

Identified Risk	Risk Severity	Proactive Mitigation Strategy
Media Latency & Unauthorized Direct Linking	High	Store videos in private Azure Blob containers; serve media strictly via time-limited Shared Access Signature (SAS) tokens; implement client-side caching to optimize bandwidth.
Concurrency & Inconsistent Enrollment States	High	Encapsulate multi-course purchase, enrollment records, and payment events inside transactional Unit of Work boundaries with explicit database isolation levels.
Credential & Certificate Forgery	Medium	Stamp each generated PDF certificate with a globally unique verification identifier (GUID), a SHA-256 integrity hash, and an embedded public verification QR code.
Resource Bottlenecks from Video Uploads	Medium	Utilize chunked, direct-to-cloud browser uploads to Azure Blob Storage, bypassing web application server memory and preventing thread exhaustion.
Scope Creep & Schedule Deviation	Medium	Adhere to agile sprints with daily standups, track deliverables through a clear issue board, and prioritize must-have capabilities across core grading milestones.

1.5 Key Performance Indicators (KPIs)

- **Platform Availability & Uptime:** Minimum 99.9% uptime verified across Azure App Service monitoring.
- **Server Response Time (TTFB):** Average server response under 800ms for database queries and under 1.2s for rendered views.

- **Video Playback Startup Time:** Initial video buffering and playback launch under 1.5 seconds.
- **Payment Processing Reliability:** 100% transactional consistency with zero orphaned order records during payment callbacks.
- **System Scalability:** Capable of handling 500 concurrent active learners without degradation in database response times.
- **Code Health & Test Coverage:** Minimum 80% test coverage across the Business Logic Layer (EduHub.Services) with zero high-severity code vulnerabilities